

Laxmi Devi Institute of Engineering & Technology

Pre University Exam (P.U.E.) 2025-26

B. Tech. I Semester, All Branches

Subject: Engineering Mathematics-I

Maximum Marks-70

Time : 3hr

Section-A

Sec-A is compulsory and each part carry equal marks. (10 × 2 = 20)

Q.1.(i) For a curve defined by the equation $x = f(t)$ and $y = \phi(t)$ Find the radius of curvature.

(ii) Define multiple point and double point.

(iii) State Eulers theorem.

(iv) The function $u = 3x^2 - y^2 + x^3$ is maximum at.

(v) Write the form of Bernoullis Differentiation equation of the first order.

(vi) Find the C.F.of given differential equation $\frac{d^2 y}{dx^2} + a^2 y = \cos ec ax$.

(vii) Write the condition of exact for second order differential equation.

(viii) Write the second order differential equation.

(ix) Find the complete integral of following differential equation.

$$z = px + qy - 2\sqrt{pq}$$

(x) Write the Charpit's Method formula:

Section-B

Attempt any 5 question out of 7. (5 × 4 = 20)

Q.1. Trace the curve $x^3 + y^3 = 3xy$.

Q.2. Find the asymptotes of the following curve: $y^3 - xy^2 - x^2y + x^3 + x^2 - y^2 - 1 = 0$

Q.3. If $u = \sec^{-1}\left(\frac{x^3 + y^3}{x + y}\right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$

Q.4. If $u = f(r)$, where $r^2 = x^2 + y^2$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

Q.5. Solve the following differential equations:

$$(1 + y^2)dx = (\tan^{-1} y - x)dy$$

Q.6. Solve the following differential equation.

$$\frac{d^2 y}{dx^2} + (3 \sin x - \cot x) \frac{dy}{dx} + 2y \sin^2 x = e^{-\cos x} \sin^2 x$$

Q.7. Solve the following differential equation.

$$\sqrt{p} + \sqrt{q} = x + y$$

Section-C

Attempt any 3 question out of 5. ($3 \times 10 = 30$)

Q.1. If CP, CD be a pair of conjugate semi-diameters of an ellipse. Prove that the radius of curvature ρ is $\frac{CD^3}{ab}$, where a and b are the length of the semi-axes of ellipse.

Q.2. In a plane triangle ABC, find the maximum value of $\cos A \cos B \cos C$.

Q.3. Solve : $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$

Q.4. Solve the following differential equation by the method of Variation of parameter: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$

Q.5. Solve the following differential equations by Charpit's Method:

$$2xz - px^2 - 2qxy + pq^2 = 0$$

12/12/25

Roll No..

Total Page No. : 2

B.TECH. 1 SEM (NEW SCHEME) MAIN/BACK

Pre - University EXAMINATION 2025-26

COMPUTER FUNDAMENTALS & PROGRAMMING

IFY2—08 - Common to All Branch

Part - A: Short Answer Type Questions (up to 25 words) 10 x 2 marks = 20 marks. All 10 questions are compulsory.

Part - B: Analytical/Problem solving questions 5 x 4 marks = 20 marks. Candidates have to answer four questions out of 7.

Part - C: Descriptive/Analytical Problem-Solving questions 3 x 10 marks = 30 marks. Candidates have to answer 3 questions out of 5. Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Part—A

1. Explain the concept of Arrays in C.
2. What is a pseudo code?
3. State the differences between Assembly and Machine languages.
4. What is literals?
5. Explain the concept of break statement.
6. What is the use of Enum?
7. Explain firmware.
8. Discuss on ternary operators.
9. Differentiate between static and dynamic memory allocation?
10. Write the notations of flow chart.

Part—B

1. What are the storage classes? Give the classification of various storage classes used in C.
2. What is 2 — D Array? Write a program to multiply two matrices A and B of size 3x3 each and store result in matrix C. Explain the difference between structure and union.
3. Convert the following numbers:
 - i. $(7EB9)_{16} = ()_{10} = ()_8$
 - ii. $(1056)_{16} = ()_8 = ()_2$
 - iii. $(5C6)_{16} = ()_{10} = ()_8$
 - iv. $(74524)_8 = ()_2 = ()_{10}$
4. What is Stored Program Architecture of computer ! Draw the architecture block diagram. Briefly describe each component and their Interconnections.

5. What is recursion? Write a C program to find the sum of N natural numbers using recursion.
6. Write the difference b/w primary memory and secondary memory.
7. What is the difference between Application software and System software.

Part - C

1. What is file? Why do we need to store data in files? write note on the following functions. give a program code that demonstrate its usage.
 - (a) fopen()
 - (b) fclose()
 - (c) fgetc()
 - (d) fputc()
2. Explain the concept of structures used in C. Write a C program to create students mark sheet (student's name, roll number and marks of 10 subjects) for 100 students. The program should be able to calculate the subject-wise and student-wise totals.
3. What are the different modes in which a file can be opened in C? Explain with suitable example.
4. Explain the concept of files in C programming. Write a program to read data from the keyboard, write it to a file called INPUT, again read the same data from the INPUT file, and display the output.
5. What is a far pointer in C? How is it different from near pointer? Explain with an example.

Laxmi Devi Institute of Engineering and Technology, Chikani (Alwar)

Department of Electrical Engineering

Pre- University Exam 2025-26

Program:- B.Tech.

Subject & Code:-BEE (1FY2-07)

Session:- 2025-26

Max. Time:-3 Hours

Branch &Semester:-All(ISEM)

Max. Marks:-70

Instruction to candidate:

- **Part-A: Shorty Answer Question (up to 25 words) $10 \times 2 = 20$ marks. All 10 Question are compulsory.**
- **Part - B: Analytical / problem solving question $5 \times 4 = 20$ marks. Candidates have to answer 5 questions out of 7.**
- **Part-C: Descriptive/ Analytical/problem solving question $3 \times 10 = 30$ marks. Candidates have to answer 3 questions out of 5.**

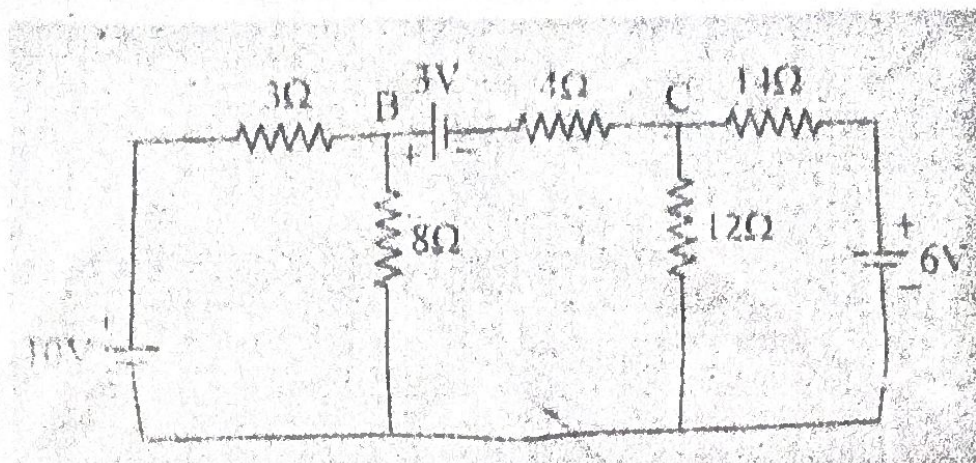
Part-A

- Q.1 Explain the principal of thevenin in electric circuit. (2)
- Q.2 Differentiate between real power, reactive power and apparent power. (2)
- Q3 Draw R and L current phasor diagram respectively (2)
- Q.4 what are the primary losses in transformer (2)
- Q.5 Explain the current source and voltage transformer (2)
- Q.6 Explain Commutator working of in DC motor. (2)
- Q.7 Draw V-I characteristics of P-N junction diode. (2)
- Q8 Write the EMF equation of transformer. (2)
- Q.9 what is SCR. (2)
- Q.10 what is modulation in communication system. (2)

Part-B

- Q.1 Explain nodal analysis with the help of example. (4)
- Q.2 what are the construction and working principal of transformer. (4)
- Q.3 Briefly discusses the DC machine with the help of diagram. (4)
- Q.4 what is TRIAC? Sketch V-I characteristics of thyristor. (4)
- Q.5 Discuss the amplitude modulation in detail. (4)
- Q.6 Explain the operation and V-I characteristics of uni-junction transistor (UJT). (4)

Q7. For the circuit shown in figure determine the voltages at node B and C. Also calculate the current through the 3Ω , 8Ω , 12Ω and 14Ω resistors. (4)



Part-C

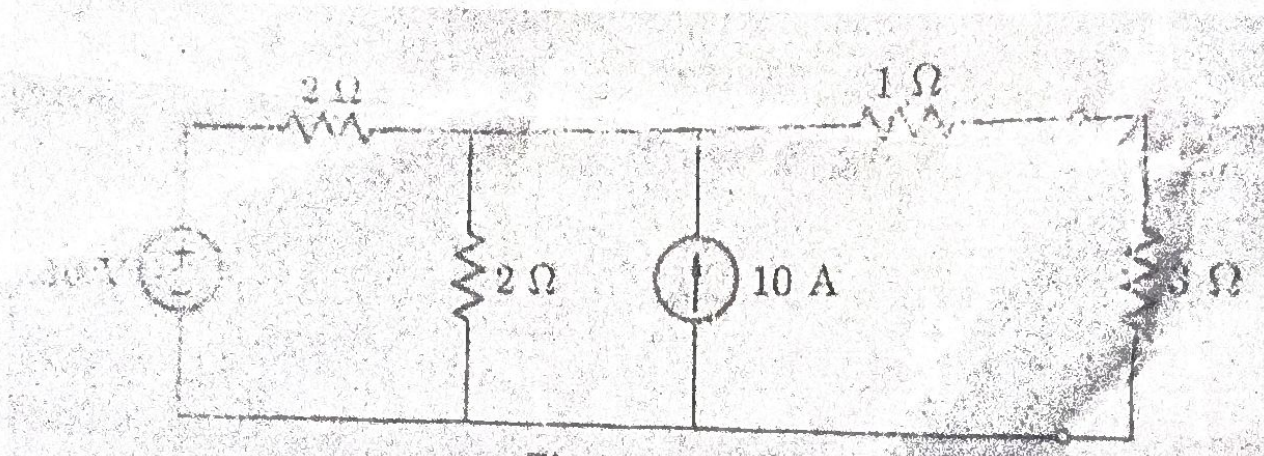
Q.1 A single phase transformer has 500 primary turns and 100 secondary turns. If the primary voltage is 200 V, determine the secondary voltage and current, assuming an ideal transformer and load of 5Ω on the secondary side.

Q.2 In the network as shown in figure determine the current through the 3Ω resistor by application of

(A) Superposition theorem

(B) thevenin's theorem

(5+5)



Q.3 Explain two wattmeter to measuring the power of 3-phase ac circuit (10)

Q.4 Draw the input and output characteristics of CE configuration (10)

Q5 Explain optical fiber, satellite and multiplexing in communication in detail (10)

Instructions to Candidates:

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Part—B: Analytical/Problem Solving questions 5×4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part—C: Descriptive/Analytical Problem Solving/Design questions 3×10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Part-A

Note: - Attempt all Questions

$10 \times 2 = 20$ Marks

1. Scope of managerial economics
2. Limitations Of Managerial economics
3. Define law of demand
4. Types of elasticity of demand
5. Define joint stock
6. Define perfect competition
7. Discuss Gross Profit
8. Define margin of safety
9. Discuss double entry system
10. Define opportunity cost

Part-B

Note: - Attempt any 5 questions.

$5 \times 4 = 20$ Marks

1. Discuss the importance of managerial economics in present Scenario.
2. Describe concept of inflation in detail.
3. Explain the factors affecting law of demand.
4. Discuss explicit and implicit costs in detail.
5. Discuss limitation of Break-even point.
6. Discuss phase of a business cycle.
7. From the following figures, calculate net profit: -

Particulars	Amount (₹)
Operating Profit	5,26,600
Interest on Long-term Loan	44,000
Loss on Sale of Furniture	70,000
Loss by Fire	50,200
Gain on Sale of Machinery	1,00,000
Dividend Received	5,600

Part-C

(3*10=30Marks)

Note: - Attempt any Three questions.

1. Explain the features of sole proprietor with their advantages and disadvantages
2. Enter the following transactions in the Journal of Manohar Lal & Sons :-

2019		₹
March 1	Manohar Lal & Sons started business with cash	60,000
2	Purchased furniture for cash	10,000
4	Purchased goods for cash	25,000
5	Bought goods from Kamlesh	15,000
10	Paid cash to Kamlesh	15,000
16	Purchased goods from Sohan	6,000
18	Purchased goods from Sohan for cash	8,000
20	Paid rent for the office	1,000

3. Explain the following: -

1. Oligopoly
2. Cross elasticity of supply
3. Partnership
4. Economies of scale
5. Cobb-Douglas production function

4. What are liquidity ratios? Discuss the importance of current and liquid ratio.

5. Show the effect of the following transactions on the Accounting Equation:

- (i) Started business with cash ₹ 50,000.
- (ii) salaries paid ₹ 2,000.
- (iii) Wages Outstanding ₹ 200.
- (iv) Interest due but not paid ₹ 100.
- (v) Rent paid in advance ₹ 150.

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